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$$\begin{aligned}
 & \int \frac{dx}{1 + \sin x} \\
 \tan \frac{x}{2} = t & \Rightarrow x = 2 \arctan t \Rightarrow dx = \frac{2dt}{1+t^2}, \sin x = \frac{2t}{1+t^2} \\
 \Rightarrow \int \frac{\frac{2}{1+t^2}}{1 + \frac{2t}{1+t^2}} dt &= \int \frac{\frac{2}{1+t^2}}{\frac{(1+t^2)+2t}{1+t^2}} dt \\
 = \int \frac{2}{(1+t)^2} d(1+t) &= \int \frac{2}{u^2} du = -\frac{2}{u} + c = -\frac{2}{1+t} + c \\
 = -\frac{2}{1 + \tan \frac{x}{2}} + c
 \end{aligned}$$

References